6. Find the number of zeros of the following polynomials represented by their graphs.

(i)

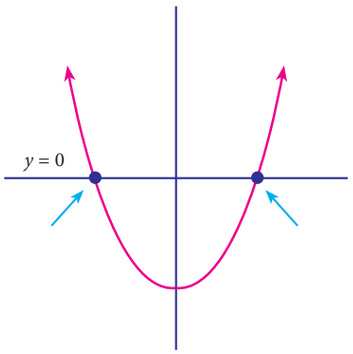


Fig. 3.10

(ii)

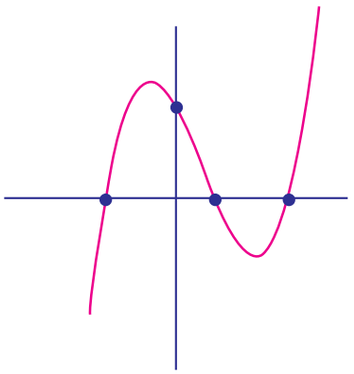


Fig. 3.11

(iii)

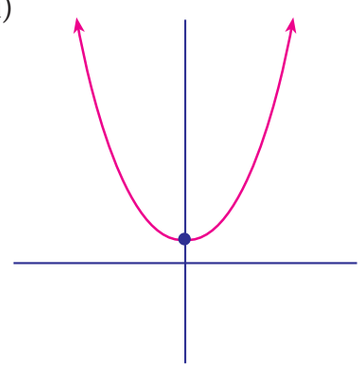


Fig. 3.12

(iv)

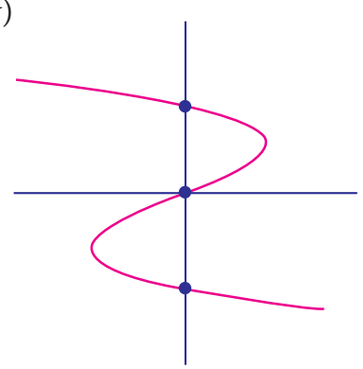


Fig. 3.13

(v)

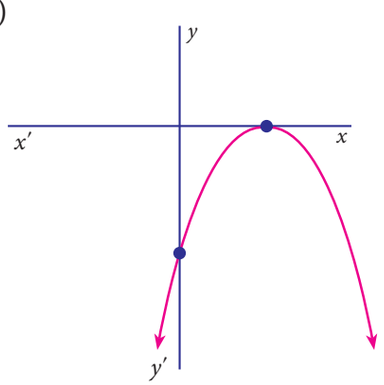


Fig. 3.14

3.3 Remainder Theorem

In this section , we shall study a simple and an elegant method of finding the remainder.

In the case of divisibility of a polynomial by a linear polynomial we use a well known theorem called **Remainder Theorem.**

If a polynomial $p(x)$ of degree greater than or equal to one is divided by a linear polynomial $(x–a)$ then the remainder is $p(a)$, where a is any real number.

Significance of Remainder theorem : It enables us to find the remainder without actually following the cumbersome process of long division.

Note

(i) If $p(x)$ is divided by $(x+a)$, then the remainder is $p(-a)$

(ii) If $p(x)$ is divided by $(ax–b)$, then the remainder is $p(\frac{b}{a})$

(iii) If $p(x)$ is divided by $(ax+b)$, then the remainder is $p(-\frac{b}{a})$

3 Algebra_Term1.indd 92 02/12/2022 21:38:55